

Project Title
Product search and Recommendation system for shopping
Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Online shopping platforms contain large product catalogs, making it difficult for users to find relevant products when queries contain spelling mistakes, incomplete terms, or multiple keywords. This project presents a **Product Search and Recommendation System for Shopping** that applies advanced Data Structures and Algorithms to improve product retrieval. The system uses a researched corpus of 20 product documents containing structured product and search information. **Knuth–Morris–Pratt (KMP)** and **Rabin–Karp** algorithms are implemented for efficient string matching and performance comparison. **Levenshtein Edit Distance** enables typo-tolerant search, while a **Trie** provides efficient prefix-based autocomplete suggestions. **HashMap** supports fast product lookup, and a **Priority Queue (Heap)** ranks products according to their relevance. A multi-stage search pipeline combines these techniques to provide accurate and efficient retrieval. The system also supports relevance-based product recommendations, demonstrating the practical application of advanced DSA concepts to a real-world e-commerce search problem.

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